



MILLIMETER-WAVE PACKAGING DESIGN FOR MACH-ZEHNDER MODULATORS

PHOTONICS · MILLIMETER-WAVE PACKAGING · 3D EM SIMULATION

CAPABILITY & TOOLS:

3D finite-element EM simulation in **Ansys HFSS** (and Keysight EMPro) · waveguide, lumped and sheet port selection · adaptive mesh refinement to an S-parameter convergence criterion · scripted parameter sweeps via solver API · T-matrix de-embedding and re-embedding · coplanar-to-microstrip transition and taper design · ribbon-bond and solder interconnect modelling · thin-film termination sizing · optical bench and hermetic package integration.

EXECUTIVE SUMMARY:

An electro-optic polymer Mach-Zehnder modulator die can carry usable modulation past 100 GHz. A package can take all of that away. Between the driver and the optical waveguide sit a coaxial connector, a solder joint, an interposer chip, a ribbon bond and a termination — five discontinuities, each invisible at 10 GHz and each significant at 110 GHz, where a 300 μm bond wire is a circuit element rather than a connection.

This paper sets out how that packaging problem is framed, modelled and solved: what the RF transitions are, how they are represented in a 3D finite-element solver, which port and meshing choices change the answer, and how T-matrix de-embedding turns a multi-hour solve into a tractable optimisation loop. It closes with the package architecture the electrical design has to live inside.

KEY RESULTS:

- **Five RF discontinuities** identified and modelled between driver and waveguide — connector, solder, interposer, ribbon bond, termination.
- **Continuous impedance taper** from a 130 μm connector pin to a 100 μm modulator pad, avoiding a stepped transition.
- **Internal thin-film termination** removes the output connector, cuts part count and frees the housing layout.
- **T-matrix de-embedding** reduces optimisation iterations from hours to tens of minutes each.
- **Launch position is a first-order variable**, not a detail — the stub it creates dominates high-frequency reflection.

INTRODUCTION:

Data-centre traffic is driving optical interconnects from 400 Gbit Ethernet toward 800 Gbit and 1.6 Tbit per link, and the modulator sets the ceiling. Silicon Mach-Zehnder modulators modulate by carrier depletion rather than a linear electro-optic effect, which caps them near 40 GHz; bulk lithium niobate reaches comparable bandwidth and is difficult to thin further. Poled electro-optic polymer on sol-gel silica avoids both limits, because its low dielectric dispersion keeps the RF and optical waves velocity-matched well past 100 GHz.

That advantage belongs to the die, not to a product. A modulator characterised on a probe station has no connectors, no housing and no fibre. Making it something a systems engineer can buy means adding a coaxial interface, an interposer, bond wires, a termination, an optical bench and fibre — and every one of those is a place where a 50 Ω line stops being 50 Ω . Packaging is where high-bandwidth modulators are won or lost, and it is an electromagnetic design problem before it is a mechanical one.

TEST SETUP:

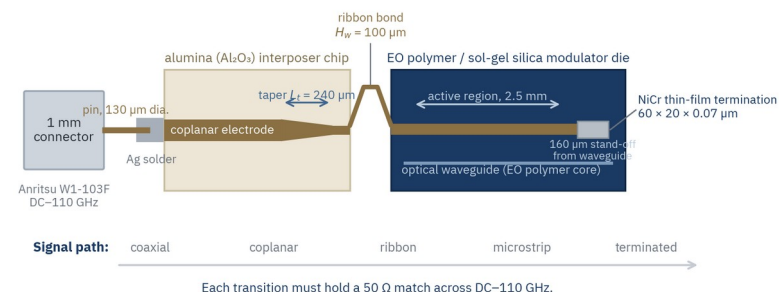


FIG 1: The millimetre-wave signal chain. Drive signal enters coaxially, transitions to a tapered coplanar line on an alumina interposer, crosses to the die by ribbon bond, runs as microstrip over the active region, and is absorbed on-chip.

Framing the model. The path is modelled end to end rather than as isolated parts, because the discontinuities interact: reflections from the ribbon bond and from the taper add or cancel depending on the line length between them. The alumina interposer's coplanar electrode tapers continuously from the connector-pin diameter down to the modulator input pad so impedance never steps. Material properties are entered as measured values rather than datasheet defaults — loss tangent in particular drives the result at these frequencies.

Port and mesh choices change the answer. Waveguide ports are used on boundary faces where the structure is driven externally, because they excite the true modal field and introduce no port parasitic inductance. Sheet ports are used

inside the solution volume where a lumped excitation is physically appropriate. Choosing the wrong one produces a plausible-looking S-parameter set that is quietly wrong. Meshing is refined adaptively against an S-parameter error criterion so that the solver concentrates elements where fields are strong — at the taper, the bond feet and the termination — rather than uniformly across a large air volume.

Making the optimisation converge. A package of this kind has of order nine interconnect variables: ribbon height and foot positions, taper length, chip metallisation thickness, solder lengths fore and aft, pin insertion depth, and the straight run ahead of the taper. Sweeping them needs hundreds of solves, and a full-module solve costs hours. The practical answer is to solve the full module once, convert S to T matrices, de-embed the portion that is not changing, and re-embed only the region under study on each iteration. Cost per iteration drops from hours to tens of minutes, which is the difference between optimising nine variables and guessing at three.

RESULTS:

Launch position is a first-order variable. Moving the feed point along the electrode changes high-frequency return loss far more than its size suggests: the open length between launch and electrode end forms a stub supporting multiple reflections, and the taper needs a straight run ahead of it. Feeding at the electrode end is the correct default, and it should be settled before any other interconnect variable is touched.

Terminate on-chip, not through a second connector. An external termination demands a second connector facing the drive connector; at millimetre-wave pitch the two bodies crowd each other and can couple. A thin-film resistor inside the microstrip line removes a connector, cuts part count and frees the housing layout from any connector-spacing constraint. It must be set back from the optical waveguide, because it dissipates the full drive power and thermo-optic coupling into the guide would distort the signal.

Match the resistor to the line, not to the nominal. A thin-film termination can outperform an idealised $50\ \Omega$ load in simulation, which looks wrong until the cause is clear: the ideal load has no thickness and so carries extra parasitic inductance, and the microstrip ahead of the termination is rarely exactly $50\ \Omega$. Optimising against the line the resistor actually sees — weighted toward the top of the band, where reflection grows fastest — beats matching to a nominal impedance.

Know what the trade costs. Lengthening the active electrode lowers half-wave voltage and drive power but costs insertion loss and therefore bandwidth, leaving return loss essentially unchanged — a loss-driven trade, so effort belongs on conductor and dielectric loss rather than further impedance tuning.

APPLICATIONS:

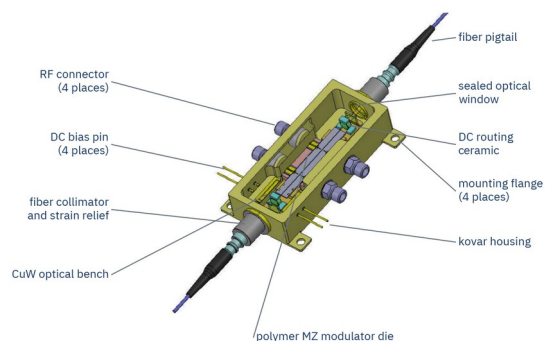


FIG 2: Package architecture. The optical train sits on a thermally stable CuW bench inside a hermetic kovar housing, with RF connectors, DC bias pins and a sealed window passing through the walls.

The RF design does not exist on its own. It shares a sealed housing with an optical train — collimators, lenses and polarizers aligned to a few microns on a thermally stable bench — plus DC bias routing, feedthroughs and fibre strain relief. Connector placement, bias-pin spacing and bench length are simultaneously electrical, optical and mechanical constraints, which is why the packaging engineer must hold all three at once.

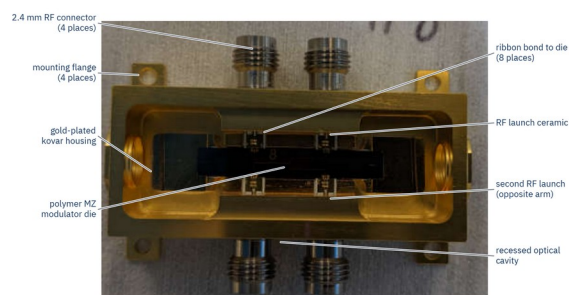


FIG 3: The design carried into hardware: gold-plated kovar housing, die mounted and ribbon-bonded to the RF launch ceramics on both arms, connectors fitted, before the optical train and lid are installed.

Figure 3 shows that architecture built. Moving from a converged model to a working assembly is where the tolerances declared in simulation get tested: ribbon length and loop height, solder geometry and die placement must be held to what the model assumed, or the optimisation is academic. The capability transfers to any high-frequency assembly where a small interconnect region governs a large, expensive-to-solve model — coherent transceivers, millimetre-wave photonics, phased-array feeds and broadband instrumentation.

DISCUSS THIS WORK:

This paper summarises one project. If the same problem appears in your programme, I am happy to walk through the method, the data behind it, and how it would map onto your hardware. emincherif@gmail.com